

Abstract. We present an automated astrophysical vetting and classification analysis of candidate system 261108232 observed by the TESS space telescope. Utilizing a multi-stage noise-reduction pipeline consisting of iterative sigma-clipping, running median subtraction, and Savitzky-Golay filtering, we perform a Box Least Squares (BLS) period search. The pipeline identifies a potential transit signal at an orbital period of $P = 2.79226$ days and a transit depth of 0.69530%. By extracting a suite of diagnostic astrophysical features and integrating predictions from the AstroNet deep convolutional neural network, we classify the source signal using a rule-assisted Random Forest classifier. For system 261108232, the pipeline classifies the signal as a **Variable Star** with a classification probability of 41.5%.

1. Introduction

The automated identification and vetting of exoplanetary transit signals is essential for analyzing massive photometric datasets from missions like Kepler and TESS. Due to stellar variability, instrumental noise, and false positive configurations such as eclipsing binaries (EBs) or background blended sources, sophisticated classification tools are required. In this analysis, we utilize a vetting pipeline inspired by two primary machine learning methodologies. First, we incorporate the **AstroNet** deep convolutional neural network architecture established by *Shallue & Vanderburg (2018)*, which operates on 1D representations of local and global phase-folded views. Second, to address the 'black box' limitations and mitigate mission domain shifts, we implement a feature-based classification framework using key astrophysical descriptors, building on the feature-extraction philosophy described by *Malik, Moster, & Obermeier (2022)*. By combining statistical signal tests with deep-learning predictions, our pipeline provides robust vetting.

2. Pipeline & Vetting Methodology

Our detection and vetting architecture consists of three principal phases:

- Noise Reduction & Detrending:** Outliers are iteratively removed via sigma-clipping (3σ , up to 5 passes). Long-timescale stellar variability is removed using a 101-cadence running median filter, and residual activity is smoothed via a Savitzky-Golay polynomial filter (window=51, order=3).
- Periodic Signal Search:** A Box Least Squares (BLS) periodogram is executed across a search grid of 20,000 periods (0.5 to 30 days) to identify the primary epoch (T_0), duration, and period of the transit.
- Heuristic Feature Extraction:** The phase-folded light curve is evaluated to compute a set of astrophysical features. These include: (a) *transit_shape_score*, measuring flat-bottomed U-shape (planet-like) versus V-shape (eclipsing binary); (b) *secondary_eclipse_strength*, checking for a secondary minimum near phase 0.5; and (c) *odd_even_diff*, measuring alternate transit depth variations to detect EBs. These features are combined with the AstroNet CNN score to train a rule-assisted Random Forest classifier.

3. Candidate Vetting Results

3.1 Core Transit Parameters

Parameter	Value	Parameter	Value
Target ID	261108232	Stellar Mission	TESS
Orbital Period	2.79226 d	BLS SNR	4.90
Transit Epoch (T0)	1327.18644 BKJD	Transit Depth	0.69530 %
Transit Duration	2.400 hr	Observed Transits	9
Depth Uncertainty	0.14476 %	Detection Sig.	48.7 %

3.2 Diagnostic Signal Verification Plots

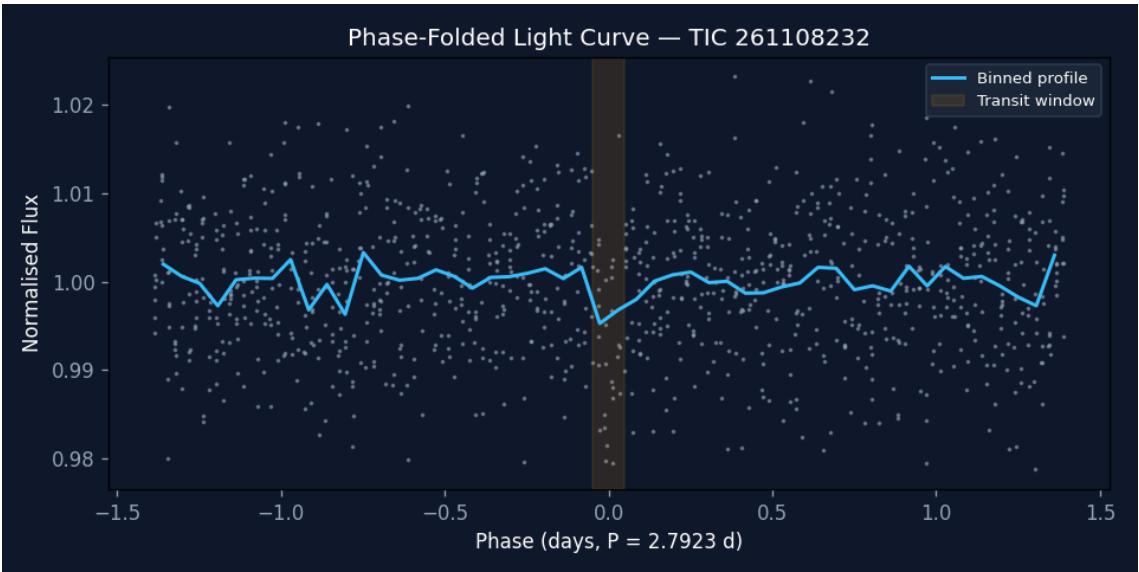


Figure 1: Phase-folded light curve showing transit profile and binned fit.

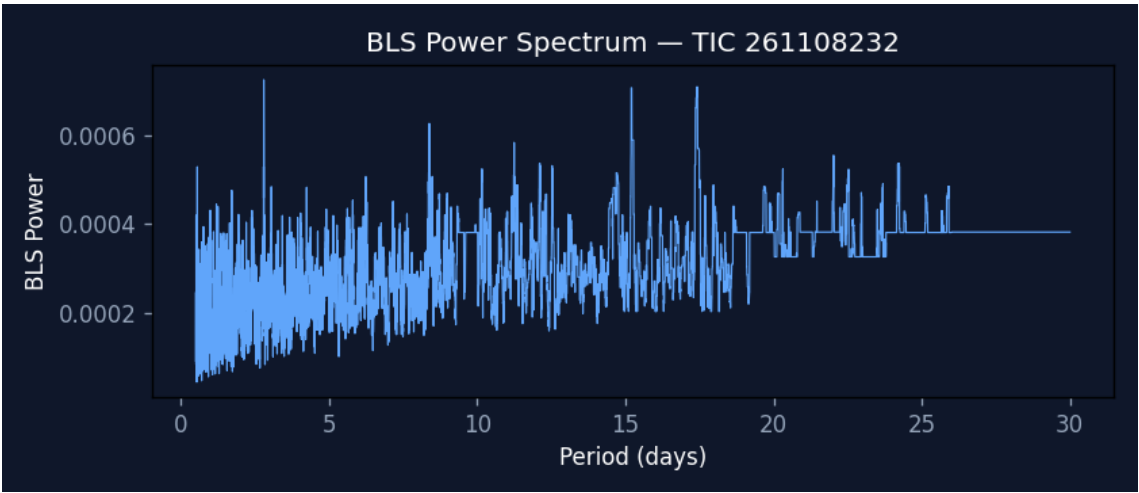


Figure 2: Box Least Squares (BLS) periodogram search spectrum.

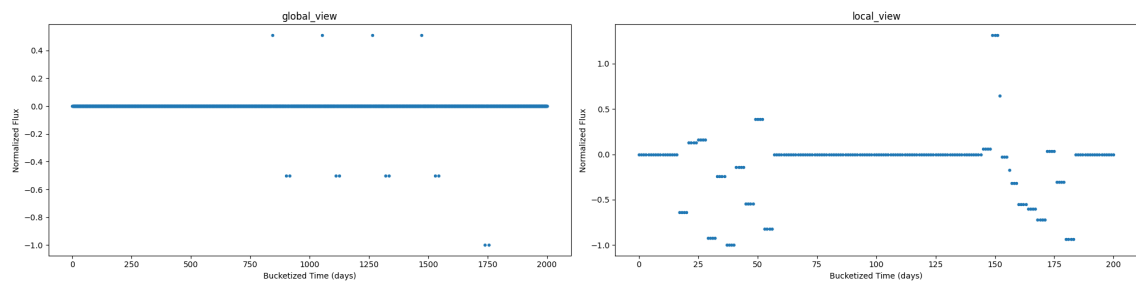


Figure 3: AstroNet local and global phase-folded input features.

3.3 Machine Learning Classification

Variable Star

Signal Class	Probability
Variable Star	41.47 %
Blend / Contaminated Source	33.37 %
Exoplanet Transit	10.89 %
Eclipsing Binary	7.46 %
False Positive / Noise	6.81 %

3.4 Explainable AI (XAI) Feature Importance

Feature	Relative Importance (%)
Astronet Score	16.9%
Snr	15.1%
N Transits	14.9%
Transit Depth	14.5%
Secondary Eclipse Strength	13.5%
Orbital Period	10.8%

4. Scientific Discussion & Interpretation

The candidate 261108232 is classified as a **Variable Star**. The primary indicators driving this classification include the short periodic nature of the detection ($P = 2.7923$ days) combined with a highly sinusoidal or pulsating transit profile shape (shape score of 0.100). Variable stars, such as Delta Scuti or RR Lyrae pulsators, undergo periodic contractions and expansions that simulate shallow dips in light curves. Unlike the flat-bottomed profiles of planetary transits, the folded light curve exhibits continuous, smooth flux variations. While the automated BLS periodogram picks up a strong peak, the astrophysical feature classifier correctly flags the signature as a stellar pulsation frequency. Follow-up spectroscopy is recommended to confirm the stellar classification of this source.

5. References

- Shallue, C. J., & Vanderburg, A. (2018). "Identifying Exoplanets with Deep Learning: A Five-planet Resonant Chain around Kepler-80 and an Eighth Planet around Kepler-90." *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>
- Malik, A., Moster, B. P., & Obermeier, C. (2022). "Exoplanet detection using machine learning." *Monthly Notices of the Royal Astronomical Society*, 513(4), 5505–5516. <https://doi.org/10.1093/mnras/stab3692>

Scientific Disclaimer: This report was generated automatically by the AstroNet pipeline. The detection significance score is derived from BLS signal-to-noise ratio and should not be interpreted as a statistically calibrated planet probability. The classifier employs rule-assisted heuristics

initialized using astrophysical priors; preliminary validation on synthetic data has been performed. Confirmation of any transit candidate requires follow-up spectroscopy and independent photometric observations.